FIG 01. FULL RA-MedFiLM-BARR Canonical Methodology & Execution Pipeline

Risk-Averse Medical Feature-wise Linear Modulation with Boundary-Aware Residual Refinement | Staged Freeze Architecture (4.84M Parameters)

PANEL A: Multi-Domain Input

RGB — ISIC 2018 (Dermoscopy)
Cutaneous Pigmented Lesion
Task t = 0

US — BUSI-WHU (Ultrasound)
Breast Benign & Malignant Mass
Task t = 1

Endo — Kvasir-SEG (Colonoscopy)
Gastrointestinal Mucosal Polyp
Task t = 2

XR — KUB-StoneX (Radiography)
Kidney, Ureteric & Bladder Stones
Task t = 3

Binary Task-Specific Formulation

- Target: 1 foreground class per task
- Input: x ∈ ℝ^(1×3×512×512), t ∈ {0,1,2,3}
- Learned task embedding: e_t ∈ ℝ^32
- G: GlobalMedFiLM (3.52M params)
- Total Assembly: 4.84M params

Task Embedding Generator E(t)
nn.Embedding(4, 32) → e_t ∈ ℝ^32
Feeds 5 FiLM Modulation Sites in G:
[bridge, dec4, dec3, dec2, dec1]

x

t, t

FORCED_SKIP

PANEL B: FULL RA-MedFiLM-BARR — Proposed Inference Architecture

STAGED COMPOSITION (0 TRAIN PARAMS)

G — GlobalMedFiLM Backbone (3.52M)

Encoder: 4 Residual Stages
Channels: [24, 48, 96, 192] (ConvNormAct + ResSkip)

Bridge (256-d) + FiLM Site 1
F̂ = (1 + tanh(γ(e_t))) ⊙ F + tanh(β(e_t))

Decoder: 4 Stages + FiLM Sites 2–5
TranspConv Up + Cat Skip + ResBlock + FiLM
dec4 (192), dec3 (96), dec2 (48), dec1 (24)

Output Head (1×1 Conv → Sigmoid)
Produces Global Probability p_global ∈ [0, 1]^(512×512)

Single Shared Output Head | 0 Head Swapping

SKIP (U < τ)

p_global

B_phi — Benefit Router (36.8K Params)

Input: 202 Numeric Features + 16 Task Emb = 218-D
• Global context, proposal area, border entropy, variance

Linear(218→128) → LN → SiLU → Dropout(0.10)
→ Linear(128→64) → LN → SiLU → Linear(64→1)

Predicts: Δ̂_final (Counterfactual Paired Gain)

Utility: U = Δ̂_final - λ·C (λ = 1/4, C = 1051648/10270749)
Decision Rule: EXECUTE iff U ≥ τ_C8_final (~ -0.024358)

If U < τ_C8_final → SKIP_GLOBAL (outputs p_global)
If U ≥ τ_C8_final → EXECUTE REFINE (proceeds to R_psi)

ROI Features

EXECUTE (U ≥ τ)

R_psi — Regional Refiner (1.28M Params)

Input: B × 4 × 256 × 256 = [RGB, p_global_crop]
Dedicated crop resolution; bilinear resize to 256×256

Narrowed U-Net: Channels [16, 32, 64, 96, 128]
4 Residual Stages + 1 Bridge | Output: 1 Channel Logit

Residual Formula: p_cand_crop = σ(logit(p_crop) + Δl_roi)
Spatial Fusion: Insert crop into canvas (NO alpha blending)
Deterministic Reco

ACCEPT (P(safe) ≥ 0.50)

REJECT (P(safe) < 0.50)

p_cand

S_omega — Post-Safety Gate (1.22K Params)

Input: 16 Numeric Features + 4 Task One-Hot = 20-D
• Global vs candidate divergence, border shift, area change

Linear(20→32) → SiLU → Linear(32→16)
→ SiLU → Linear(16→1) → Sigmoid Logit

Evaluates Safety Probability: P(Δ > 0) = σ(logit)

Acceptance Rule: ACCEPT iff P(safety) ≥ 0.50
Threshold: τ_safe = 0.50 (Fail-Closed Governance)

P(safety) ≥ 0.50 → EXECUTE_THEN_ACCEPT (Output p_cand)
P(safety) < 0.50 → EXECUTE_THEN_REJECT (Fallback p_global)
Rejection discards harmful crop updates, preventing degradation

PANEL C: Staged Training Strategy

STAGE 1: Global Backbone Training (Phase 21)

- Target: GlobalMedFiLM (G) + Task Embedding E(t)
- Dataset: TRAIN5832 (ISIC, BUSI, KVASIR, KUB)
- Loss: $\mathscr{L}$_global = $\mathscr{L}$_BCE + $\mathscr{L}$_Dice on 512×512
- **FiLM init at zero: initial forward is pure identity**

→ **Checkpoint FROZEN before Stage 2**

STAGE 2: Regional Refiner Training (Phase 25A)

- Target: BARRRegionalRefinerV1 (R_psi, 1.28M)
- Input: 4-Channel crops [RGB, p_global_crop] (256×256)
- Loss: $\mathscr{L}$_refine = $\mathscr{L}$_BCE_boundary + $\mathscr{L}$_Dice
- G remains completely FROZEN (no backprop to G)

→ **Checkpoint FROZEN before Stage 3**

STAGE 3: Counterfactual Router Fit (Phase 25D)

- Target: BARRBenefitRouterV1 (B_phi, 36.8K)
- Ground Truth: Out-of-fold paired Dice gain: Δ_Dice
- Loss: $\mathscr{L}$_router = MSE(Δ̂_pred, Δ_true) + λ_cost·Cost
- 5-Fold Cross-Fit on ROUTER_DEVELOPMENT (n=4,957)

→ **Checkpoint FROZEN before Stage 4**

STAGE 4: Post-Safety Classifier Fit (Phase 25F)

- Target: BARRPostSafetyV1 (S_omega, 1.22K)
- Label: Binary y_safe = 𝕀(Δ_Dice > 0)
- Loss: $\mathscr{L}$_safety = Binary Cross-Entropy (BCE)
- Trains on comparative divergence features (20-D)

→ **All 4 modules frozen for Phase 26 Assembly**

No Joint Training / No End-to-End Backprop

- Each module trained on frozen upstream outputs
- Assembly wrapper introduces exactly 0 new params
- Guarantees perfect determinism & provenance
- **Avoids catastrophic gradient interference**

INSET 1: Detailed FiLM Feature-Wise Linear Modulation Mechanism in GlobalMedFiLM

Mathematical Formulation:

Given task embedding e_t = E(t) ∈ ℝ^32, affine conditioning generators compute scale γ and shift β:

$$\gamma(e_t) = \tanh(W_\gamma \cdot e_t + b_\gamma) \in (-1, 1), \beta(e_t) = \tanh(W_\beta \cdot e_t + b_\beta) \in (-1, 1)$$

Applied to feature map F ∈ ℝ^(C × H × W) at 5 key decoder/bridge locations:

$$\hat{F}_(c, h, w) = (1 + \gamma_c(e_t)) \cdot F_(c, h, w) + \beta_c(e_t)$$

- Zero-Initialization: W_γ, W_β initialized to 0 → γ = 0, β = 0 at epoch 0 (exact identity mapping F̂ = F).
- Non-Invasive: task_id is never concatenated as an image channel or separate output head.

INSET 2: Regional Residual Refinement & Spatial Canvas Fusion Logic

Residual Refinement Formulation:

Cropped patch x_roi concatenated with prior p_roi ∈ ℝ^(4 × 256 × 256) fed into R_psi:

$$\Delta l_roi = R_psi([x_roi, p_roi]),\ p_cand_crop = \sigma(\text{logit}(p_roi) + \Delta l_roi)$$

Deterministic Recomposition (Spatial Insertion without alpha blending):

$$p_fused(x, y) = p_cand_crop(x - x0, y - y0) \text{ if } (x, y) \in \text{Window; else } p_global(x, y)$$

- Bounded Correction: The residual refiner only adjusts ambiguous boundary contours, preventing core shifts.
- Fail-Closed Protection: If S_omega detects uncertainty, p_fused is discarded and p_global is restored.

• Expansion: 1.50× scaling centered at (cy, cx)

Figure 1 Caption (Official IEEE TMI / MedIA Format):

Figure 1. Canonical methodology of FULL RA-MedFiLM-BARR. (A) Heterogeneous multi-domain inputs (dermoscopy, ultrasound, endoscopy, radiography) mapped to discrete task embeddings e_t. (B) Core inference architecture: GlobalMedFiLM (G, 3.52M params) generates initial probability p_global; the Benefit Router (B_phi, 36.8K params) evaluates expected paired gain U = Δ̂ - λ·C, routing candidates or triggering SKIP_GLOBAL; Regional Refin